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COLLECTIVE DECISION-MAKING WITH STUBBORN AGENTS

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SUMMARY

Situated at the intersection of psychology and artificial intelligence, this thesis explores the phenomenon of collective decision-making, central to societal, political, and global issues. Drawing from the concepts of swarm intelligence observed in social insects and its applications in fields like robotics, this research investigates the role of stubborn artificial agents, or "zealots", in an online human swarm experiment. While some mathematical models suggest that zealots can improve decision-making, real-world results often reveal a more complex scenario, including the potential for group division. Employing a digital paradigm, the current experiment asked participants to perform a forced-choice task. The study comprised 178 online trials, each of two minutes duration, across five types of groups. These groups exhibited varying proportions of zealots, four ranging from 20% to 80%, and one comparison group with no zealots. The results revealed that as the number of zealots increased, the group became more divided, and the correct majority opinion - defined as the proportion of participants and artificial agents holding the same correct opinion - decreased. However, the effect of zealots on the participants majority - defined as the proportion of participants holding the same opinion regardless of its accuracy – was not significant. Additionally, zealots only had a mild negative impact on the accuracy. This study points to possibly fruitful future research that could involve improved online tools, larger groups of participants and additional factors such as participants' confidence in their decisions to better understand the role of artificial agents in collective decision-making.

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1. GENERAL INTRODUCTION

At the intersection of neurosciences, psychology, and artificial intelligence lies a phenomenon that binds together creatures as disparate as ants, humans, and robots: the capacity for collective decision-making. This ability is defined as a sub-field of collective behaviour that focuses on how groups make decisions without centralized leadership (Bose et al. (10)). The aim of this research is to examine collective decision-making within an online human swarm experiment that incorporates artificial agents. It seems relevant to study collective decision-making in the context of human civilization as it is central to political and democratic processes and, essential for addressing global issues like climate change or the intricacies of vaccination campaigns. Hence, our collective decision-making abilities significantly influence the trajectory of societies and the planet.

Social insects, such as ants and bees, offer some of the most sophisticated examples of collective decision making in the natural world, a subject of study since the publication of the significant book 'Animalibus Insectis Libri Septem' in 1604 (Nicoli Aldini et al. (45)). They live in colonies that can number from a few thousand individuals to millions, and they must make collective decisions about a variety of issues critical to their survival. These decisions involve finding the shortest route to a food source among thousands of possible paths, deciding where to establish a colony, and determining how to distribute tasks among workers. The emergent collective behavior exhibited by a group of social insects is commonly referred to as "swarm intelligence" (Bonabeau et al. (78)); however, it is worth considering that the term 'intelligence' might be somewhat specific and may not encompass cases where the collective behavior is maladaptive. To understand swarm intelligence, one must acknowledge that the intricacy of a single insect is insufficient in describing the complexity of what social insect colonies can achieve. The most challenging question is perhaps how to link individual behavior to group performance (Bonabeau et al. (9)). Such explorations have led to efficient applications for humans in various fields. For instance, Southwest Airlines collaborated with researchers who developed algorithms inspired by the foraging behavior of ant colonies to optimize cargo routing in the airline's operations. By applying these algorithms, they identified more efficient ways to load and transfer cargo, even if it meant initially sending packages in the opposite direction. The technical

implementation led to significant improvements in efficiency, slashing freight transfer rates by as much as 80% at the busiest cargo stations, decreasing the workload for cargo handlers by up to 20%, and resulting in an estimated annual gain of more than $10 million for the company (Bonabeau et al. (78)).

In recent years, these observations have also inspired new theories and empirical advancements within the field of robotics leading to swarm robotics. Initially, the field focused on cellular robotics, as introduced by Beni, Hackwood, and Wang in 1989. Later on, Bonabeau et al. (9) broadened the scope to incorporate attempts to design algorithms or distributed problem-solving devices inspired by the collective behavior of social insects and other animal societies, including humans. Krause et al. (33) tie together approaches from seemingly disparate areas by means of a general definition of swarm intelligence: “Two or more agents obtain knowledge independently, or at least partially independently, and these distinct packets of data are merged and processed through social interaction, resulting in a solution to cognitive difficulties that cannot be implemented by isolated individuals.”. Despite their robustness, scalability, and flexibility, swarm robotics systems have yet to be employed in real-world scenarios and remain limited to academic research (Dorigo et al. (64)). However, they hold great potential for various applications such as exploration, surveillance, and search and rescue.

Drawing the evolution of swarm intelligence within collective decision-making underscores a transition that has been addressed below - from insect studies, in 1604 to the advent of robotics in 1989. Situated between these points in time, a parallel concept emerged in 1906 : Collective Intelligence introduced by Sir Francis Galton's experiment. Galton asked a crowd at a county fair in England to estimate the weight of an ox (Rosenberg et al. (79)). The crowd was made up of a mix of farmers, butchers, and regular fairgoers, none of whom were experts in estimating the weight of an ox. Galton's goal was to demonstrate that the average voter was not very smart. However, he was shocked to find that the average of the 787 estimates (out of 800) was nearly perfect, even though the vast majority of individual estimates were way off. This led to the birth of the field of Collective Intelligence, which focuses on tapping into the wisdom of crowds to make better decisions.

This work lies at the intersection of these concepts, examining their interplay and potential synergies among them. Indeed, unlike Sir Francis Galton's experiment where data is aggregated after the fact, nature uses real-time closed-loop systems with continuous feedback to enable large groups to work in synchrony and converge on solutions together. Although humans lack the innate ability to swarm that other species use to establish feedback-loops among members, new technologies have enabled groups of online human users to form real-time closed-loop systems modelled after natural swarms: Human Swarming (Rosenberg et al. (79)). Transitioning from physical to virtual environments enables the investigation of human groups within simulated systems. These systems extend beyond the conventional wisdom of the crowd, integrating procedures that allow the group to function as a cohesive entity. This leads to the emergence of a unique property - human swarming, in turn contributing to a more nuanced understanding of collective decision-making.

In this context, the study's focus is on introducing stubborn artificial agents into an online human swarm experiment. These agents, named zealots, are programmed to never change their opinions. While primarily mathematical and artificial models have identified positive effects of implementing zealots (Vendeville et al. (59), Prasetyo et al. (49)), when human and social interactions are considered, outcomes tend to be more nuanced and can potentially lead to polarization or backlash effects (Jones et al. (66)). The negative effect of implementing zealots in human groups could be due to irrational behaviors inspired by loyalty to the group or fear, among other factors (Jones et al. (66)). On the other hand, the presence of zealots could increase diversity of opinion (Grag et al. (80)). When individuals with differing perspectives and opinions come together, they can more effectively evaluate and analyse a problem, ultimately leading to better decisions (Grag et al. (80)). Through a literature review, this research will address the key points highlighted in this introduction. It will then provide a detailed description of the experimental process, including hypotheses and the results, aiming to contribute to a better understanding of the advantages and disadvantages of incorporating stubborn agents in collective decision making. Ultimately, this study will conclude with a discussion of the findings and potential directions for future research enhancement.

2. REVIEW OF LITTERATURE

This literature review chapter serves as a foundation for the empirical research that follows. By examining the literature, we hope to contribute to the ongoing conversation in the field and to identify opportunities for future research that can advance our understanding of collective decision making. The chapter will provide an overview of the current state of knowledge and explore the underlying mechanisms that drive group decision-making.

This chapter is organised into four main sections that span diverse fields ranging from entomology to robotics, all guided by the common theme of collective decision-making. First is a brief overview of decision making in non-human animal groups with a focus on swarm intelligence in social insects. The second section of the literature review will focus on robotics, which has been partly inspired by swarm intelligence in social insects. The various approaches and techniques that have been developed to achieve swarm robotics will be explored, such as different robot algorithms strategy and voting models, eventually leading to a key milestone in the field of artificial agents: the development of agents that exhibit unwavering opinions, known as zealots.

In addition to artificial agents, the literature review also encompasses human groups. Therefore, in the third section, we will investigate human-human swarms. Reviewing the existing literature on the factors that influence collective decision-making in groups of people and exploring parallels and differences between human and robotic swarms. Finally, the last section aims to integrate the examined frameworks to investigate how the presence of zealots can both hinder and facilitate the achievement of a consensus in human groups.

3.1. DECISION MAKING IN NON-HUMAN ANIMAL GROUPS

Most non-human case studies of Swarm Intelligence, particularly in invertebrates, are based on the belief that the individual animal is cognitively simple and restricted, whereas the group is capable of more remarkable achievements (Krause et al. (33)). Couzin et al. (15) reveal that, the larger the group, the smaller the proportion of informed individuals needed to guide the group toward a goal, and only a very small proportion of informed individuals is required to achieve great accuracy.

Some animals have an innate predisposition to migrate in a certain direction (Berthold et al. (5), Berthold et al. (6)), individuals can respond to abiotic cues such as thermal gradients that may aid migration (Neil et al. (44), Grunbaum et al. (23)) and experienced group members can often play an important role in guiding those who are less experienced or inexperienced (Couzin et al. (15)). A minority of informed individuals within schools of fish are known to be able to influence the foraging movement of the group and help the group navigate towards a target (Reebs et al. (51)).

3.1.1 "Weighting model" for group on the move

Similarly, it only requires very few individuals (approximately 5%) within honeybee swarms to guide the group to a new nest site (Seely et al. (52)). On another note, group members of large insect swarms or schools of fish do not systematically have the ability for individual recognition (Reebs et al. (51)). Couzin et al. (15) address two major issues in the absence of signal mechanisms: "how information about the location of resources, or of a migration route, can be transferred within groups; and secondly, how individuals can achieve a consensus when informed individuals differ in their preferences ?". To answer those questions, Couzin et al. (15) created a model where informed individuals balance the influence of their preferred direction and their social interactions with a weighting term ("w").

They proposed the following equation:

$$\mathrm{d}_i{}'(t+\Delta t) = \frac{\hat{\mathrm{d}}_i(t+\Delta t) + \omega \mathrm{g}_i}{|\hat{\mathrm{d}}_i(t+\Delta t) + \omega \mathrm{g}_i|}$$

Where d_i represent a desired direction of travel and "g_i" the preferred direction (Couzin et al. (15)). As "w" approaches 1, individuals tend to balance their preference to move in a certain direction while maintaining social interactions with group members and if "w" exceeds 1, individuals are more heavily influenced by their own preferred direction than by their neighbours (Couzin et al. (15)). They found that the accuracy of group motion (in a preferred direction) for a given group size increased asymptotically as the proportion of informed individuals increased (Couzin et al. (15)). Hence, the

larger the group, the smaller the proportion of informed individuals needed to guide the group with a given accuracy (Couzin et al. (15)). They expect colonies to have evolved to reach the asymptote of accuracy and they predict that having a larger scout population would not achieve a greater benefit (Couzin et al. (15)).

3.1.2 Value-based decision

Decisions are mainly considered in terms of accuracy, but animals frequently face equally viable alternative solutions. For example, a bee finds two potential nest sites available, both of equal quality. If the quality of the options available is low, the bee should wait until the next opportunity. If the quality of both is high, they should choose as quickly as possible (Bose et al. (10)). This is called a value-based decision, in contrast with an accuracy-based decision. A key prediction of what is called the "value-sensitivity hypothesis" is that choices between equally-valued options will proceed faster when the options have a high value than when they have a low value (Pirrone et al. (47)). This had led to a speed-value trade-off rather than a speed-accuracy trade-off where, as a consequence, accuracy in decisions over small differences is sacrificed in favour of quick responses (Pironne et al. (47)). Pironne et al. (47) reported similar data about humans and monkeys. They "hypothesize that far from being an artefact of imperfect implementation, longer RTs with low-value alternatives and shorter RTs with high-value alternatives are diagnostic of an adaptive decision strategy for the uncertain environments faced by decision making systems, at different level of biological complexity, and in various domains."

The next section of the thesis explores the realm of robot swarms, examining how the principles of decision-making observed in biological swarms can be applied to robotic systems. By understanding and implementing these principles, sciences can harness the power of swarm intelligence to develop more efficient, robust, and adaptable robotic systems for various applications, ranging from environmental monitoring to search and rescue missions.

3.2. ROBOT SWARM

A robot swarm, also known as swarm robotics, refers to a large group of decentralized, self-organized robots that work collectively to achieve a specific task or goal (Beni et al. (62). Swarm robotics is inspired by the behaviour of social insects, such as ants, bees, and termites, which exhibit swarm intelligence (Bonabeau et al. (63)) as explored in the session above. Swarm robotics has gained significant attention in the scientific community due to its potential applications in various fields such as, agriculture, environmental monitoring, search and rescue operations, and transportation (Dorigo et al. (64). Hence, to be efficient, collective decision-making strategies should not only possess fast and accuracy but also a level of generality that allows them to be transferred and applied to diverse problem domains (Brambilla et al. (12)). This section explores robot strategies implemented in robots in order to explore them further. Additionally, the central notion of this thesis will be introduced and developed: stubborn artificial agents called “zealots”.

3.2.1 Collective perception in a robot swarm

Valentini et al. (56) propose a novel problem scenario known as "collective perception" to explore and evaluate the performance of swarm robot decision-making strategies. The primary goal of their study is to enable a swarm of robots to efficiently explore their environment and evaluate the frequency of certain features that are scattered across it (e.g., the availability of precious metals; the presence of pollutants or cancer cells) with the objective to determine which feature is the most frequent. The researchers implemented the collective perception scenario in a swarm robotics system composed of 20 e-pucks. The e-puck is a commercial robot designed for research and education. It has a diameter of 7 cm, an autonomy of 45 minutes and it can move at a maximum speed of 16 cm/s with wheels. In its basic configuration, the robot is equipped with RGB light-emitting diodes, a bass camera, an accelerometer, a sound sensor and 8 proximity sensors. Fig. 1 illustrates the robotic platform and the experimental setup (Valentini et al. (56)).

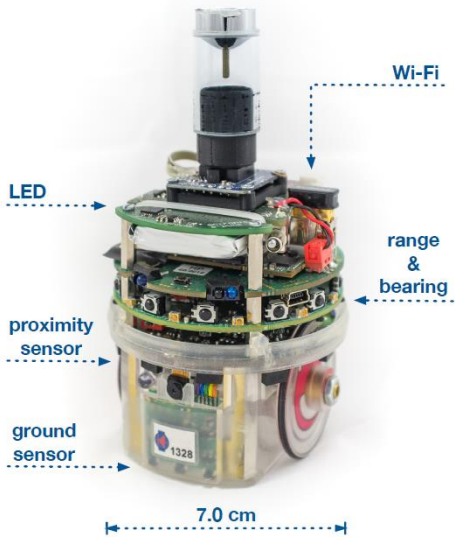

Fig. 1 (left): the e-puck robot with details of its hardware.

Fig. 2 (right): top-view picture of the collective perception scenario with two features represented by the black and white colours.

To achieve this objective, the authors compare three different swarm robotics decision-making strategies: Direct Majority Decision-Making (DMMD), Direct Majority with Variable Decision threshold (DMVD), and Distributed Consensus (DC). These strategies are employed to address the challenges faced by the swarm of robots, such as communication constraints, scalability, and adaptability (Valentini et al. (56)).

DMMD is a simple majority decision-making mechanism, where each robot adopts the opinion that is broadcasted by the majority of observed neighbours. In contrast, the DMVD approach employs the voter model as its strategy, where a robot takes on the opinion of a randomly selected neighbor. Additionally, the DC approach lets robots compare their quality estimates directly to modify their opinion. Those three strategies are characterized by a common structure of the robot controller that is implemented as a probabilistic finite-state machine (PFSM) (Valentini et al. (56)).

The PFSM consist of two controlled states: the exploration and the dissemination state. In the exploration state, a robot will explore the environment by performing a random walk and his routine of obstacle avoidance while estimating the quality of the

environment. The exploration state has a random duration. In dissemination, robots share their opinions and mix positions to facilitate decision-making. The duration of dissemination is modulated based on the quality estimate, thereby promoting the spread of the best opinion.

The nuanced findings of this study reflect the complexity of robotic science and revealed that DMMD and DMVD offer adaptable solutions for decision-making scenarios beyond their original site-selection context while maintaining speed and accuracy with a promising level of generality. However, they demonstrated the advantages of a self-organized approach by showcasing the scalability issues of the DC strategy's consensus time. DC is more robust to noise although it relies on less information. In certain problem configurations, the DC strategy surpasses both DMMD and DMVD strategies in performance. Nevertheless, the communication demands of this approach necessitate the use of sufficiently advanced robots (Valentini et al. (56)).

3.2.2 Stubborn agents named ”zealots”

While swarm robotics harnesses the power of collective intelligence to accomplish complex tasks, the introduction of stubborn agents, or "zealots," into these systems is an opportunity to explore their impact on group dynamics and decision-making processes. Recent empirical research (Bonabeau et al. (9), Dorigo et al. (17), Helbing et al. (25), Coucke et el. (14)) studied the effects of injecting stubborn and noisy agents in natural or artificial groups and networks. The term “zealots” is borrowed from a religious Jewish group from the first century. They literally interpreted a tradition of "zeal" that had been present in the Bible for centuries that urged the strengthening of the covenant between God and Israel, involving violence and radicalism (Mézange et al. (39)). In the 17th Century, the word “zèlote” appears in the dictionary referring to someone animated by religious fanaticism (Mézange et al. (39)). More recently, it has been adopted to characterize a natural or artificial intelligence that does not change opinion or choice.

3.2.3 Zealots in artificial data

The effect of a zealot in a cognitive or computer science context is mainly observed in random subjects' groups although there may be value in seeing how it affects more homogenous groups. Various models have been proposed to address the topic of when and how quickly a population can reach a consensus (Vendeville et al. (59)). Mukhopadhyay et al. (43) also demonstrated that with more than 25% of zealots, if one viewpoint is initially favoured, consensus is established with a probability that converges to 1 as the network size grows. Vendeville et al. (59) created a general model strategy with artificial data for shifting internal diversity of opinion toward a desired target level based on an arbitrary degree distribution network with a path from any user to another (Vendeville et al. (59)). The key feature is the introduction of zealots who do not modify their opinion, forming an inflexible core of agents that have immense persuasive power (Vendeville et al. (59)). Researchers (Vendeville et al. (59)) simulated an example of a full graph at various times and show that if the zealots are present in only one camp, then the entire population will eventually adopt the opinion of the zealots' camp. On the other hand, if Zealots are present on both sides, this prevents consensus then the system reaches a state of equilibrium where no opinion prevails in which it fluctuates indefinitely around the ratio: Zealots 1 divided by Zealots 0 plus Zealots 1 (Z1 /(Z0 +Z1)).

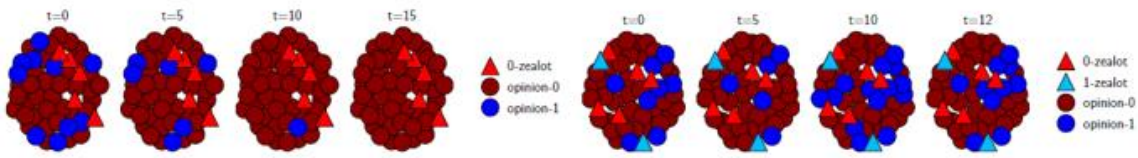

Fig. 3: Left: Example realisation of the model on a complete graph at different times with n = 50, n1 = 10; Z0 = 5, Z1 = 0. Because there are no 1-zealots, everyone eventually adopts opinion 0. Right: same setting except z1 = 2. Because there are no zealots in botch camps, the system reaches a state of equilibrium where no opinion prevails.

3.3.4 Reaching better accuracy with Zealots in a best-of-n problem

Prasetyo et al. (49) performed a study of the best-of-n problem in dynamic environments with two alternatives available (n=2). Valentini et al. (57) describe the best-of-n problem as requiring a swarm of robots to make a collective decision over which option, out of "n" available options, provides the best solution to encounter the occurring needs of the swarm. They use the term options to abstract domain-specific concepts associated with various application scenarios (e.g., foraging patches, aggregation areas, traveling paths) (Hunter et al. (28)). Given a swarm of N robots, they say that the swarm has found a solution to a particular instance of the best-of-n problem as soon as it makes a collective decision with a large majority of robots favouring the same option (Hunter et al. (28)). As often in this investigation field, the best-of-n problem takes biological inspiration from the collective behaviour of social insects such as ants and particularly bees (Bradley et al. (11)).

3.3.4.1 Voter model in a dynamic environment

Prasetyo et al. (49) they investigate a swarm of robots with limited capabilities that could only interact locally to reach consensus on the option with the best quality from two options (n = 2). Robots were in a dynamic environment, meaning the quality of the options would change over time. More precisely, the difference in quality between the two options remained constant for some time and then was swapped in order to explore how agents would adapt.

The robots were considered to be able to measure quality, while the environment was symmetric in terms of the distribution of the n possibilities, implying that all options could be assessed in the same amount of time on average. Robots could show one alternative at a time, the one that corresponds to their present position, and employ a decision mechanism to modify their choice after observing other robots in their neighbourhood. The decision mechanism they used is the voter model, the most well-known decision mechanisms used in the swarm robotics literature ((Baronchelli and Díaz-Guilera (4), Valentini et al. (58)). The voter model allows agents distributed over a network to interact only with their neighbours. At each round, a randomly picked agent adopts the opinion of a random neighbour. The evolution of the process

continues until consensus is eventually reached (Valentini et al. (58)). An advanced voter model, referred to as the weighted voter model, has been proposed by Valentini and al. (58), whereby for a time proportional to the quality of the option corresponding to their opinion, robots broadcast their choice to other robots within their range of sight, a concept that is inspired by Honeybees' waggle dance habit (Prasetyo et al. (49)).

Half of the agents were initialized with a state corresponding to option A and the other half with option B state (Prasetyo et al. (49)). After leaving the nest, the authors explored and measured the quality of the site associated with their opinion, during an exponentially distributed amount of time per agent that did not depend on quality or option quality (to avoid clustering on their way back to the nest) (Prasetyo et al. (49)). At the end of this exploring time, the agents switched subsequently from exploration state (with opinion A or B) to dissemination state with their new option state (if they changed) and travelled back to the nest at a different time (Prasetyo et al. (49)). During the dissemination phase, agents were performing random walks and locally broadcasting them opinion continuously to other robots around this time. The duration while they were disseminating was correlated to the site quality they had last visited, with the consequence of being more likely to connect with the best quality option and to make the group converge toward it using the voter model (Prasetyo et al. (49)).

3.3.4.2 Results

Among the large set of features Prasetyo et al. (49) manipulated, there may be relevance in exploring the results of different combination of features, such as the ratio of quality options (i.e. the difficulty with which the better option can be identified) defined as high or low, the swarm size and the proportion of stubborn agents. The main outcome of the study was that without stubborn individuals, the group did not adapt to a dynamic environment for low and high value ratios. With stubborn individuals, a consensus was achieved when the quality ratio was high, while the proportion of stubborn individuals did not play a significant role. In the case of low quality ratio, the adaptation was very poor. Additionally, increasing the proportion of stubborn

individuals reduced response time but in a non-linear effect with a rapid saturation (Prasetyo et al. 49)).

Nevertheless, if a large number of stubborn agents can have no effect or detrimental effects (fixed proportion of Zealots tested: 5% and 20%), according to the researchers, keeping their number at a bare minimum fixed number can achieve a good level of consensus (Prasetyo et al. (49)). Indeed, with a fixed level of stubborn agents (S=10), results were still good in a high quality ratio and increased to reach a good level in low quality ratio as long as the swarm size was large enough (Prasetyo et al. (49)). The simulation did not converge for N=100 but did for N= 1000 and 10.000.

Recent research has shown that the introduction of stubborn agents, or "zealots," into these systems can impact group dynamics and decision-making processes. Studies have explored the effects of injecting stubborn and noisy agents into natural or artificial groups and networks, showing that the presence of zealots can prevent or promote consensus depending on various factors such as the quality ratio and swarm size. Additionally, research has shown that incorporating stubborn agents into the best-of-n problem can improve the swarm's ability to adapt to dynamic environments and achieve consensus. However, the optimal number of stubborn agents is still subject to debate, with studies showing that a fixed minimum number can achieve good levels of consensus while a large number can have detrimental effects. This literature review has revealed that the presence of stubborn agents or "zealots" can significantly impact group dynamics and decision-making processes.

3.3. “HUMAN-HUMAN SWARM”

The above discussion has mainly focused on the behaviour of robots in swarm robotics, but it is worth noting that swarm intelligence and collective decision-making are also relevant to human-human interactions. In fact, the concept of human swarms has been used to describe groups of people working together in a similar way to robot swarms, using collective intelligence to solve complex problems or achieve a common goal. Research on swarm intelligence in human groups has shown how different factors

such as group size, diversity, and individual characteristics can influence decision-making processes and group performance (Surowiecki et al. (65)).

3.3.1 The "wisdom of the crowds"

One area of focus in psychology has been determining whether a group can surpass high-performing individuals and understanding the limitations of what a group of a certain size and composition can do collectively (Krause et al. (33)).The concept of the "wisdom of crowds" suggests that a group can achieve better decisions than an individual (Bose et al. (10)). This idea is rooted in the interindividual diversity in cognitive capacity and the existence of top-performing individuals (Kern et al. (31)).

Nicolas de Condorcet in the 18th century was one of the first to highlight that individuals can benefit from collective decision-making (Sumpter et al. (54)). He postulated that the probability of a group making a correct collective decision was correlated to the proportion of individuals who were correct (Surowiecki et al. (65)). Consequently, Condorcet's theorem is primarily concerned with how consensus judgments are reached when a majority already possesses accurate information (Krause et al. (33)).

Francis Galton, in 1906, empirically established social intelligence (SI) in humans by observing visitors at a livestock fair participating in a contest to guess an ox's weight in pounds (Galton (22)). The results were surprising: while no individual, including cattle experts, could determine the ox's exact weight, the mean value of all the visitors' estimations was a mere pound more than the animal's actual weight [101]. In other words, the collective estimation was far more accurate than any individual guess. Galton had discovered that a group of people was indeed wiser than each of its individual members, including experts (Surowiecki et al. (65)). The television show Who Wants to be a Millionaire? provides another illustration of this phenomenon (Surowiecki et al. (65)). Contestants have to answer multiple-choice questions to win the prize, and they can use "lifelines" for assistance. An analysis revealed that only 65 percent of the experts were right, whereas the audience's choices were accurate in 91 percent of cases (Surowiecki et al. (65)).

However, not all groups are wise, as history has shown through instances of unruly crowds and thoughtless group behaviour. Friedrich Nietzsche once stated, "Madness is rare in individuals – but in groups, parties, nations and ages it is the rule." (Nietzsche, F. W. (77)). Errors in collective intelligence can lead to faulty decisions, but if a group can maintain diversity, collaborate effectively, and encourage open discussions, the wisdom of crowds is less a meaningless cliché but and more a potential truth. In summary, large groups have a greater capacity for problem-solving than individuals.

Both the Wisdom of Crowds and Swarm Intelligence (SI) are concepts that emphasize collective intelligence, but they differ in their mechanisms. While Wisdom of Crowds relies on external aggregation of data and each person's decision is made independently of the others, with no interaction or communication between them, SI is self-organized. Krause et al. (33) offer a comprehensive definition of SI that encompasses both animal and human phenomena: two or more individuals obtain information independently or at least partially independently, and these different packages of data are combined and processed through social interaction, leading to a solution to a cognitive problem that isolated individuals cannot achieve. SI is a strategy that people can employ to help transcend some of their own cognitive constraints.

A contemporary application of SI fitting the previous definition called “Userdriven content” is used by the LEGO company to involve LEGO users in the design of new toys. The company created an online platform in which users of Lego can talk about their design ideas, showcase their work and vote on each other’s designs (Krause et al. (33)). These debates frequently result in extremely imaginative new designs with the potential to reach the mass market and contribute to Lego's achievements. Other recent successful use of SI is observed in the collective management and prediction markets (Krause et al. (33)).

The discovery that the judgment of a diverse group can beat an expert or even a small group of experts under certain conditions (Laughlin et al. (35), Hong et al. (27)) has led to speculation that SI breakthroughs may render experts obsolete, to the point that even firms’ CEOs may be less needed in the future (Surowiecki et al. (55)). However, it is doubtful that leadership will be supplanted by SI (Kraus et al. (33)). The reason for this is that after solving a cognitive problem in swarm, a single bee or ant will

never be in possession of the overall information or solution Kraus et al. (33). On the other hand, humans can purposefully set out to apply SI principles to their benefits to gain, for instance, a competitive advantage in business (by better predicting market development). The major point is that a single expert (or team expert) can employ the entire SI mechanism (data gathering, processing, and solution). As a result, the user potential of SI in animals and humans differs significantly in this regard. In animals, SI serves as a facilitator for a group of highly interdependent individuals, and in humans, it serves as a tool to improve decision making (Kraus et al. (33)).

3.3.2 Self-organized leadership

Numerous empirical researchers have discovered that supplying people with information can turn them into leaders (Reebs et al. (50), Flack et al. (19)). However, those studies trained individuals to become "experts" and become more accurate decision makers (Freeman et al. (21)). While it is likely to find such experts in some situations, Kurvers et al. (34) highlight that in many other circumstances the individuals that have improved knowledge vary over time, depending on, for example, "context-dependent abilities and experience, time investment and orientation" (Kurvers et al. (34)). Those researchers conducted an experiment where they hypothesized that self-organized leadership correlated to information quality. This process predicts that people will contribute to a group choice based on the quality of their personal data. Individuals who have high-quality personal information are more likely to take the lead in collective decisions, whereas those who have low-quality personal information tend to follow (Kurvers et al. (34)). They tested this mechanism with a simulated predator detection task with human subjects. They briefly saw a picture of a group of animals that sometimes contained a critical amount of predators and sometimes not. Participants had to first make an individual decision whether they wanted to "stay" or "escape".

In the second part of the experiment, there was a collective decision regarding the same image, in which individual decisions were apparent to other members of the group by means of two bar charts indicating the number of people who voted "stay" or "escape". Hence, individuals could make an early decision, potentially influencing

others, or they could wait for social information. They found that individuals with poor personal information decided to wait for more social information and followed the tendency started by faster and more accurate individuals. Hence, the second decision made by the group was significantly more accurate than the first one made by individuals by themselves; meaning hierarchy can be self-organized based on individual reaction times (Kurvers et al. (34)). As we transition from the examination of self-organized leadership to the study of self-organized pedestrian interactions, we find a common thread in the adaptability of individual behaviour in response to a collective context, thereby demonstrating that both information quality and interpersonal interactions play critical roles in shaping collective decisions and movement patterns.

3.3.3 Self-organized pedestrian interactions

Understanding pedestrian interactions provides insight into collective human behaviour because these interactions embody real-time decision-making, adaptation to dynamic environments, and complex interpersonal coordination - all core components of swarm intelligence. In order to predict the dynamics that emerge at the scale of a crowd, several studies (Helbing et al. (25), Helbing et al. (24), Yu & Johansson (31)) tried to understand how pedestrians move and interact with one another. The basic modelling concept proposes that a pedestrian's motion can be explained by a combination of a driving force, which reflects the pedestrian's internal motivation to move in a specific direction at a certain desired speed, and repulsive forces, characterized by the consequences of interactions with other pedestrians and limitations such as walls or obstacles in streets (Moussaïd et al. (42)).

Moussaïd et al. (42) used a novel approach that measured the behavioural consequences of pedestrian encounters under controlled conditions. Their experimental technique tried to link both the individual and collective level of observations by assessing the interaction rules between people. They aimed to answer the question of how a walker modifies and adapts behaviour in response to interactions with other pedestrians. The mechanisms they highlighted allowed for the construction of accurate and reliable crowd models which were compared to empirical data of

pedestrian flows recorded in a crowded street. To quantify the behavioural effect of interaction, they used conditions both with and without interaction in the pedestrians' trajectory. More specifically, they observed the behaviour of an individual walking in a corridor under three distinct conditions: “1: in the absence of interactions, 2: in response to a standing pedestrian, and 3: in response to a pedestrian moving in opposite direction” (Moussaïd et al. (42)). Condition 3 had two subjects with the identical instructions as condition 1, but they had to start from opposite ends of the corridor and avoid each other while they systematically met in the centre which was the area called the “decision zone”.

The results reveal that the resulting choice of the pedestrian passing another is biased. When avoiding a stationary pedestrian, pedestrians favour the right side, but the asymmetry was much more pronounced when both pedestrians were advancing. This demonstrates how the reciprocal adjustment of two interacting pedestrians significantly accentuates the individual left/right bias. Moussaïd et al. (42) assume it could be beneficial for a person to acquire a preference that is shared by the majority of individuals. Most individuals would eventually follow the same method due to a self-reinforcing process (Moussaïd et al. (42)). Because both sides are equal at first, the theory predicts that distinct preferences arise in different parts of the world, which is exactly what happens (Helbing et al. (26)) and may be interpreted as a cultural bias (Moussaïd et al. (42)).

Amplification bias enables the crowd to become more functional and might explain the counterintuitive discovery of increased traffic efficiency in some counterflow circumstances (Algadhi et al. (1), Helbing et al. (24)). Similar amplification effects have been seen in a variety of different group-living creatures, where an individual choice is magnified by the actions of other individuals and changes the collective organization (Ame et al. (2), Bon et al. (7)). This is reminiscent of the behaviour of zealots previously examined in this section, and in this case the subjects maintained the same direction and side despite potential obstacles. It would appear that in both human and non-human natural environments, randomisation and stubbornness occur regularly in order to explore the environment and optimize goal

achievement. The expectation is that using artificial models to explore this further could help to understand and predict features of social cognition.

3.3.3 Human networks

As described above, collective behaviour is influenced by individual characteristics such as the knowledge possessed. Furthermore, it is also a matter of interpersonal connections between individuals. Theoretical research implies that structural characteristics of naturally occurring networks influence behaviour and dynamics (Kearns et al. (30). Kearns et al. (30) experiment underlines the importance of consensus reaching for collective decision-making. It supports the view that the wisdom of crowds, as well as swarm intelligence, are not simply about the accumulation of individual knowledge but also about the interactions and connections between individuals. The experiment highlights how the structure of the network and the availability of local versus global information can impact the decision-making process and the time it takes for a group to reach a consensus.

Kearns et al. (30) looked at human subject networks in an attempt to address the graph or network colouring problem The colouring problem appears on a graph constituted of lines connected to vertices; to solve it, players must colour the vertices in such a manner that no two neighbouring vertices have the same colour. This problem represents situations when it is helpful to separate one's behaviour from that of one's network neighbours. In a network of 38 vertices, individuals simultaneously control a single vertex and try to solve the complex graph colouring problem. In this challenge, the collective goal is for every player to select a colour that is different from all of their network neighbours (Kearns et al. (30)). The experiment was made with six networks divided in two families: basic cycle-based networks and networks generated by preferential attachment. Networks generated by preferential attachment are those in which vertices that are already well-connected are more likely to obtain more connections as the network is formed incrementally (Kearns et al. (30)). The likelihood of a new vertex connecting to existing vertices is not uniform, but that being linked to a vertex with a large number of connections has a bigger probability of occurrence

(Barabási et al. (3)). The chromatic number of the cycle-based networks were two (i.e. the smallest number of colours needed to colour the vertices of the graph so that no two adjacent vertices share the same colour) and three and four for the preferential attachment.

In Kearns et al. (30) experiment, collective performance was strongly affected by network structures and view fields. The networks generated by preferential attachment have shown to be far more challenging than any of the cycle-based networks. In addition, the interface permitted each participant to have a local view (their own colour and those of the neighbours). In a minority of trials, subjects were provided with a global view. Varying the locality of information highlighted strong differences between the two main categories of networks. While increasing the quantity of information (i.e. participants accessing a global view) lowered solution durations for cycle-based networks, it substantially raised solution times for preferred attachment (Barabási et al. (3)). Possible explanations suggest "information overload" related to the complexity of a network. Alternatively, it could also be that distraction made by examining more distant regions slowed down the arrival at collective solutions.

In Kearns et al. (30) study, the subjects' behaviour was mostly about attempting to avoid conflicts with closely connected neighbours. Many participants mentioned they provoked conflicts in their neighbourhood even when they had the correct colour, in an attempt to disturb the global state from a perceived stasis or local minimum. This behaviour might be compared to the intentional introduction of randomization or thermal noise into typical optimization algorithms like simulated annealing. This is also reminiscent of swarms of ants where individuals are modelled as having stochastic behaviour (Bonabeau et al. (9)) in exploring and finding information.

3.4. STUBORN AGENT IN HUMANS GROUPS

This theorical chapter began with an exploration of the concept of swarm intelligence in social insects and its influence on swarm robotics. Attention was then focused on stubborn agents, referred to as zealots, in robotic decision-making and their role in achieving consensus. In the previous section, we examined the notion of collective

wisdom, which proposes that a group of individuals can outperform a single person. The last section aims to integrate those frameworks together and investigate the impact of stubborn agents on human groups.

With a starting point for exploring stubborn behaviour in humans is examining what can cause a group to malfunction. Exploring the boundaries and limitations of a counter-example to a malfunctioning group will allow us to understand the factors that can make a group efficient. In this context, zealots can potentially both hinder and facilitate the achievement of a consensus in human groups. Through an exploration of real-world examples and virtual experiments, the aim is to provide a greater understanding of the role of zealots in human group decision-making.

3.4.1 The madness of the crowds

The earlier section entitled “The wisdom of the crowds” explored how people join forces because they understand that they can accomplish more as a team than as individuals. They make decisions together in the belief that the resulting decisions will be, on average, better than those made by any one individual (Surowiecki et al. (65)). However, there is an opposing view that too many people working together can lead to chaos and poor decision-making , e.g. “too many cooks spoil the broth” and “The madness of crowds” (Surowiecki et al. (65)). "The Madness of Crowds" is a phrase coined by Scottish journalist Charles Mackay in his 1841 book, "Extraordinary Popular Delusions and the Madness of Crowds. The book explores various historical events, such as financial bubbles, witch hunts, and religious manias, and examines how collective beliefs or behaviours can lead to irrational and destructive outcomes (Surowiecki et al. (65)). Mackay argues that the "madness of crowds" arises when individuals relinquish their critical faculties and become part of a mob mentality, fuelled by emotions such as fear and greed. The term "the madness of crowds" has since become a widely used metaphor to describe group dynamics and the potential for irrational behaviour in large groups of people (Surowiecki et al. (65)). This view that the many are unable to govern themselves, let alone make rational decisions, is probably the most widespread in political science from Plato to contemporary elitist democrats (Surowiecki et al. (65)).

The phenomenon of the "madness of crowds" manifests itself in the political arena, where collective decisions may not necessarily align with rational, individual thinking. A example can be seen in a recent Pew Research Centre study conducted on a nationally representative American Trends Panel, which highlights how voters, in the face of socio-political upheavals, stuck with their party affiliations, possibly indicating a form of collective irrationality spurred by group identity or loyalty. In the study, it was found that despite a tumultuous two-year period marked by a global pandemic, mass protests against racial injustice, and a presidential impeachment, an overwhelming majority of both Republican and Democratic voters retained their party affiliations (Jones et al. (66)). Only 9% of voters from each party changed their affiliations between 2018 and 2020 (Jones et al. (66)). The study, which involved interviews with 11,077 registered voters on five occasions between September 2018 and July 2020, revealed that there was no significant net shift in the overall balance of party identification in the electorate (Jones et al. (66)). The fact that overwhelming majorities of both Republican and Democratic voters retained their party affiliations during this tumultuous period may suggest a form of collective irrationality. Voters may be influenced by a sense of group identity or loyalty, leading them to maintain their affiliations despite significant events that could potentially change their political perspectives. A parallel can be drawn between zealotry and political party affiliations (Jones et al. (66)). The position of certain unyielding actors, akin to zealots, can strengthen the rigidity of political groups, making them more resistant to change or new ideas, reinforcing entrenched ideologies and promoting inflexible decision-making processes. As demonstrated in the numerical model based on human data by (Vendeville et al. (59)), the “back-fire” effect that occurs after injecting zealots into the role of group leaders to respond adversely and reinforce the polarization level.

Moreover, in (Sunstein et al (67)), a remarkable empirical observation reveals that group deliberation tends to push both the group and its individual members towards more extreme positions in line with their previous beliefs. For instance, individuals who initially oppose the minimum wage may become even more opposed after discussing the topic with others with the same opinion ; those who are inclined to support gun control may develop an even stronger enthusiasm for it after group discussion (Sunstein et al (67)). Hence, when a group includes a high number of strongly convinced agents, it is more likely to experience

a stagnation in the decision-making process, as the stubborn agents may resist change and amplify extreme positions. The incorporation of stubborn agent algorithms into our experimental design aims to simulate the behaviour of these intransigent agents in a simplified scenario.

3.4.2. The filter-bubble

As in politics, social media platforms are a context where polarization emerges. In recent years, social media platform recommendation algorithms have brought together people with the same opinion while substantially amplifying confirmation bias by displaying materials to users that is most likely to fit their preferences - the so-called "filter bubble" effect (Eli Pariser (18)). Confirmation bias is a cognitive bias that leads individuals to favour information that confirms their pre-existing beliefs or values while disregarding or downplaying evidence that contradicts them. People with confirmation bias may actively seek out, interpret, or remember information in a way that supports their views, leading to biased decision-making and potentially reinforcing inaccurate beliefs (Kappes et al. (68)). This “filter bubble” thesis posits that personalization algorithms on social networking sites and other platforms create personalized information ecosystems that isolate individuals from opposing viewpoints, leading to detrimental consequences for democracy (Eli Pariser (18)). As a result, increasingly isolated, close-knit online groups of like-minded individuals have emerged) in numerous sectors, such as politics (Wesley et al. (60), Michaela Del Vicario et al. (40), Kiran Garimella et al. (32)), healthcare, and research (Vendeville et al. (59)).

A primary factor contributing to the "filter bubble" phenomenon is the exponential growth in the internet (Eli Pariser (18)). During an average day, there are 900,000 blog posts published, 50 million tweets shared, 60 million Facebook updates posted, and 210 billion emails sent (Eli Pariser (18)). As the former CEO of Google, Eric Schmidt, puts it: “To record and store the two thousand years of human communication prior to 2003, one would need five billion gigabytes of space. Yet this amount of storage would only hold two days of communication in 2011.” (Eli Pariser (18)). Without personalization, the internet would be a disorganized and inefficient platform, causing us to spend hours scrolling to find a movie,

event, or music that aligns with our preferences. Platforms such as Netflix, Amazon, or iTunes would turn into inefficient and tedious experiences (Eli Pariser (18)). To cope with this information overload, people have turned to personalization, using filters and recommendation features (Eli Pariser (18)), but, according to some, this has come at the cost of our critical thinking and freedom of thought (Bozdag & van den Hoven, (69)).

However, selective exposure research shows that people generally prefer information that supports their view rather than challenging it (Dahlgren et al (70)), but the effect is not particularly large (Hart et al. (71)). People do not necessarily avoid challenging information, especially when they are confident in their beliefs or find the information useful, interesting, or curious. Even if the individual is only seeking supporting information, it may become difficult to avoid challenging information due to the size of the network, the number of sources and the number of overlapping networks of different kinds.

For Dahlgren et al (70), the important distinction between societal technological levels is not clarified by Eli Pariser (18). Firstly this means, the immediate technological situation in which any single choice affects the content recommended by personalisation algorithms and, second, the causes and consequences of these choices and technologies for humans and society. While some studies at the technological level find evidence of extremely large filter bubbles by using mathematical models or simulations (Chitra & Musco (72)), research that incorporates human and societal perspectives typically finds little to no evidence for filter bubbles, and often even discovers the opposite (Zuiderveen Borgesius et al. (73), Bruns et al (74)). The distinction can be highlighted in this thesis too. Upon closer examination, it appears that primarily mathematical and artificial models have identified positive effects of implementing Zealots (Vendeville et al. (59), Prasetyo et al. (49)). However, when human and social interactions are considered, the outcomes tend to be more negative, such as polarization or backlash effects (Jones et al. (66)).

3.4.3 Cognitive diversity

The fact that zealots can contribute to polarization as previously explored raises questions about the balance between their potential positive and negative influence. One critical notion that could tip the balance is the potential for Zealots to maintain diversity of

opinion and potentially foster cognitive diversity. However, a note of caution must be taken when drawing this parallel as the zealots in this context are artificial agents. Nonetheless, this concept provides a perspective worth further exploration. Hong and Page (75) have demonstrated that "cognitive diversity" contributes to the development of collective intelligence. Assuming that increasing the number of participants enhances this form of diversity, while keeping all other factors constant (e.g., controlling for confounding variables like communication expenses), it is probable that the collective intelligence of a larger group will be superior (Hong and Page (75)). Cognitive diversity encompasses a variety of perspectives (how situations and problems are represented), diversity interpretations (how perspectives are categorized or divided), diversity heuristics (methods for generating solutions to problems), and predictive models (the process of determining cause and effect) (Surowiecki et al. (65)). Surowiecki et al. (65) highlights that cognitive diversity is a characteristic of the group as a whole, rather than any individual member within it. For instance, in problem-solving situations, what matters most to the quality of the collective answer, is not so much how smart individuals are but how cognitively diverse they are (Hong and Page (75)).

In other contexts, however, cognitive diversity matters as much as individual ability. In such a context where judgments are aggregated to a collective answer, Hong and Page (75) account, the reason why the aggregation of predictive models can do such a great job at producing correct decisions comes from the existence of negative correlations between voters' predictions. One advantageous aspect of this type of negative correlation is that it ensures that when one person makes an error, another person is more likely to produce the correct answer, and vice versa. Consequently, these individual mistakes tend to cancel each other out in a systematically beneficial manner. This leads to the group's "average" error (i.e., the collective error in instances where calculations can be made, such as in predicting quantities) being smaller than the average error of a randomly chosen individual (Surowiecki et al. (65)).

3.4.4 The balance between madness and wisdom

Similarly, a recent study conducted in 2019 by Toyokawa et al. (76) delves into the enduring enigma of why groups of individuals occasionally display collective 'wisdom' while, at other times, exhibit maladaptive 'herding' behaviour. The researchers investigate whether and how human social learning strategies regulate the trade-off between collective intelligence and inflexible herding behaviour. The authors hypothesize that collective decision-making can generate "wisdom" or "madness" based on the specific learning strategies employed. To explore this hypothesis, they combine experimentation, theoretical modelling, and computational analysis. They conducted the experiment online with a sample of 699 persons and used hierarchical Bayesian model-fitting to identify the individual learning strategies exhibited by participants (Toyokawa et al. (76)). The study utilized a multi-player "multi-armed bandit" gambling task framework with both asocial and social sources of information.

The model they created analysed two major components: the conformity exponent (i.e. the tendency to imitate the others) and the social learning weight. Results indicated that when conformity is not too high, increasing group size can increase decision accuracy. However, as the conformity bias increases, the likelihood of rigid herding behaviour emerges. Additionally, when both parameters are substantial, instances of collective intelligence become scarce, with inflexible herd behaviour prevailing as the dominant characteristic (Toyokawa et al. (76)). In conclusion, the study demonstrates that understanding the precise learning strategies employed by individuals can predict the balance between collective wisdom and madness in group decision-making processes (Toyokawa et al. (76)).

3.5. CONCLUSION

This review of literature has explored the complex dynamics between collective intelligence and the potential pitfalls of group decision-making. While the wisdom of crowds demonstrates the power of collective decision-making, the madness of crowds highlights the potential for irrationality and chaos when individuals succumb to mob mentality. In the realm of politics, stubborn agents or Zealots can contribute to polarization

and stagnation in decision-making processes. Moreover, the phenomenon of filter bubbles on social platforms exacerbates confirmation bias, further entrenching polarized perspectives.

However, cognitive diversity offers a promising counterbalance in which a variety of perspectives, interpretations, heuristics, and predictive models contribute to the development of collective intelligence. The introduction of Zealots in this context could potentially enhance cognitive diversity by introducing unique perspectives and firmly held beliefs. Manipulating group diversity may facilitate the exploration of the impact of varying levels of collective diversity, social learning, and conformity. By understanding these complex dynamics, we can better harness the wisdom of crowds while mitigating the pitfalls associated with the madness of crowds.

3. MATERIALS AND METHODS

This research is based on an online experiment which introduced a variable number of stubborn agent algorithms to investigate the impact of such agents on the human decision-making process. Systematically grouped as a total of 10 agents per session, participants had to explore a two-colour platform (Fig. 4) with the intention of reaching a consensus about the dominant color by changing their robot's color. The 10 agents per group were composed of a variable even number of humans participant from 2 to 10 and the remaining even number of zealots from 0 to 8.

4.1 ENVIRONEMENT

The experiment took place through multi-player games on a virtual platform. Participants received a URL link to log in the platform. They entered their name and age and waited. Once the attended number of participants was stably connected, a 1-minute tutorial video began. Following the tutorial, each participant became an avatar moving randomly on a two-colour environment shown in Fig. 4.

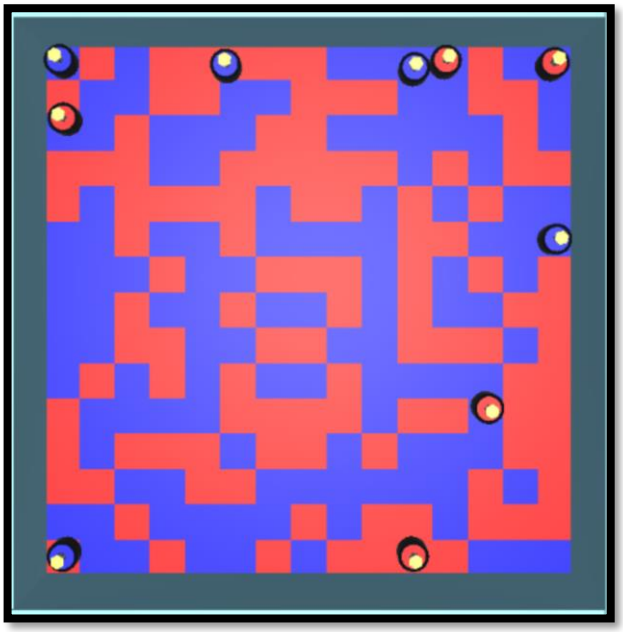

Fig. 4 : Illustration of the environments composed of 225 tiles (15x15) - Each tile is stochastically coloured (blue or red) at the beginning of each trial so that the difference between the percentage of red and blue tiles is 4% .

To build our experiment, we used HugoS ("Humans Go Swarming"). HugoS is a multi-user virtual environment tool coded with the game development soft Unity. It is used to

understand experiments in human-human swarm intelligence (Coucke et al. (14)). Existing platforms record experiment data, such as avatar locations in a virtual environment and the analyses are performed externally. HugoS allows researchers to examine recorded data internally and incorporate real-time feedback into the experiment. Participants received feedback on the screen at the end of each trial about their individual performance and the majority performance e.g. “The majority was wrong” – “You were correct”.

HugoS has various functions which can be used to explore three classes of experiments: human collective decision making, human social learning strategies, and agent-level human interaction with artificial swarms, including robot swarms (Coucke et al. (14)). This experiment concerned the third class: each session contained human’s participants and artificial agents both programmed to move randomly. Additionally, zealots were programmed to keep the same decision (colour) during the whole trial. HuGoS allows humans to play with artificial agents indistinguishable for human players as shown in Fig. 5 (Coucke et al. (14)).

Random walks proceed in a linear move for a time interval obtained from an exponential distribution with an average of 40 seconds (Coucke et al. (?)). Subsequently, the avatars rotated with an arbitrary angle evenly distributed between -180 and +180 degrees and monitor if they are within 30 meters of an obstruction (either a wall or a different robot) (Coucke et al. (14)). When detecting an obstacle, the robot determines the angle in relation to the obstacle and then rotates so its path is directed opposite to that angle (a 180-degree turn), before continuing its random walk (Coucke et al. (14)).

At the beginning of each trial, the avatars, which consist of a cylindrical base with a 7-meter diameter, are randomly assigned either red or blue colours, with an equal distribution of 50% for each color. Robots possess a consistent speed of 10 meters per second and maintain a circular field of view with a 50-meter radius surrounding their centre.

Players had a virtual camera following their movement and rotation which provides a third view of their avatar. In order to avoid too many features to analyse, we did not allow the participants to change their field of view, a functionality available with HugoS. The option allowing written communication between subjects was also avoided.

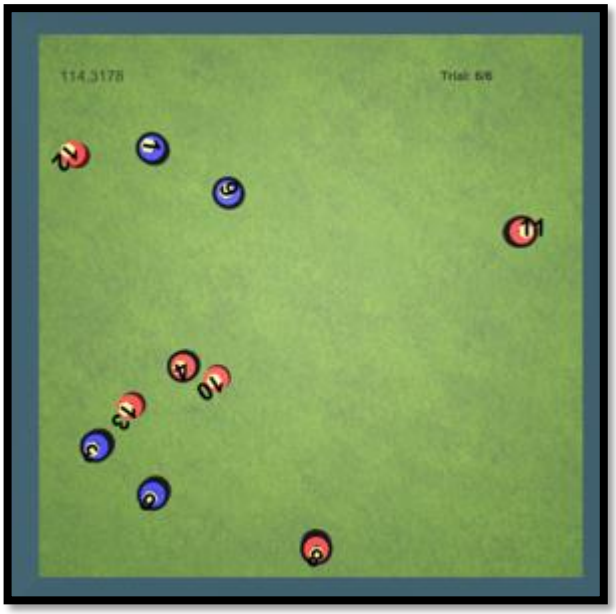

Fig. 5 : view of a trial played by 6 humans and 4 Zealots (2 blues and 2 red)

4. 2 GROUP AND TRIALS

Each group of 10 agents performed a session. To build a group of 10 agents, we combined an even number of humans from 2 to 10 and the remaining number of stubborn artificial agent (Zealots) necessary to reach 10 (see Fig. 6). The number of Zealots are also even with 50% of blue and red colour never changing. Each session contained 6 trials of 120 seconds. 30 online sessions from 5 different types of groups were performed to arrive at the 178 trials shown in Fig. 7. The total number of trials was supposed to be 36 per group but some variations occurred due to some instable connexions. The programme recorded the data for every trial in which the participants were stably connected for a minimum of 50% trial duration.

Group 1: 2 humans + 8 zealots	Group 4: 8 humans + 2 zealots
Group 2: 4 humans + 6 zealots	Group 5: 10 humans + 0 zealots
Group 3: 6 humans + 4 zealots	

Fig. 6 : The *study included* five types of groups with 10 agents

Type of Group	*Number of Participants per group*	*Number of Zealots*	*Number of Groups*	*Total Number participants*	*Number of Trial per Group*	*Total number of Trials*
1	2	8	6	12	6	38
2	4	6	6	24	6	37
3	6	4	6	36	6	35
4	8	2	6	48	6	37
5	10	0	6	60	6	32
TOTAL :				180		178

Fig. 7: summary of the quantities

4.3 DECISIONS-MAKING

Our study partially replicates a setup by Valentini et al. (56) and described further in the Robot Swarm section of the literature review. In the setup for this experiment, participants chose to either be in night mode, corresponding to Valentini's exploration phase where robots estimate the tile's color frequency, or in day mode which corresponds to Valentin's dissemination phase, where robots broadcast their opinion on which color was most frequent and adjusted their opinion after comparing it to one of their neighbors. Participants in night mode could explore the environment by seeing the tiles colours but could not see their neighbour's colour which were all black (Fig. 8). In day mode, participants could disseminate their opinion by flowing into an environment where everyone could see their neighbour's but not the tiles, as the platform was green (Fig. 9). Participants could express their choice by changing the colour of their avatar by pressing B (blue) or R (red) on the keyboard and switch from one mode to another by pressing the space bar.

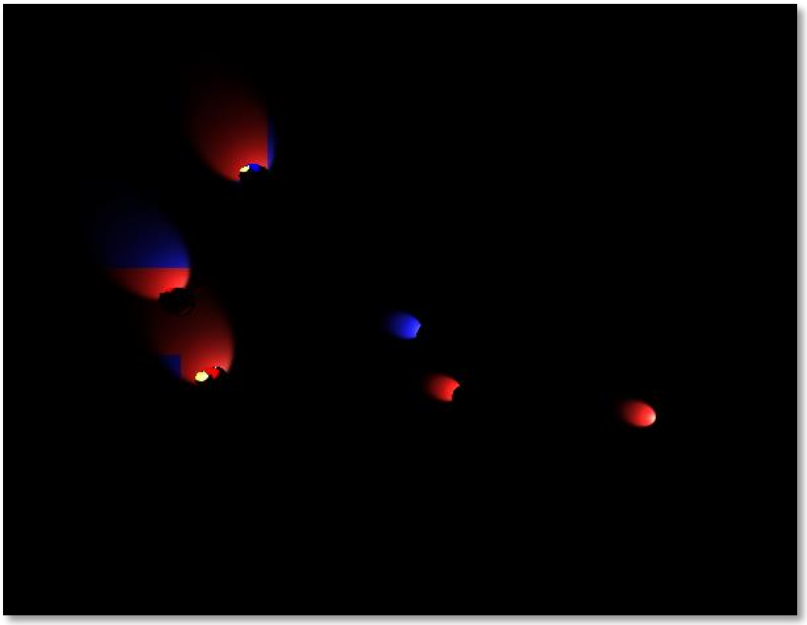

Fig. 8. : the view in night mode

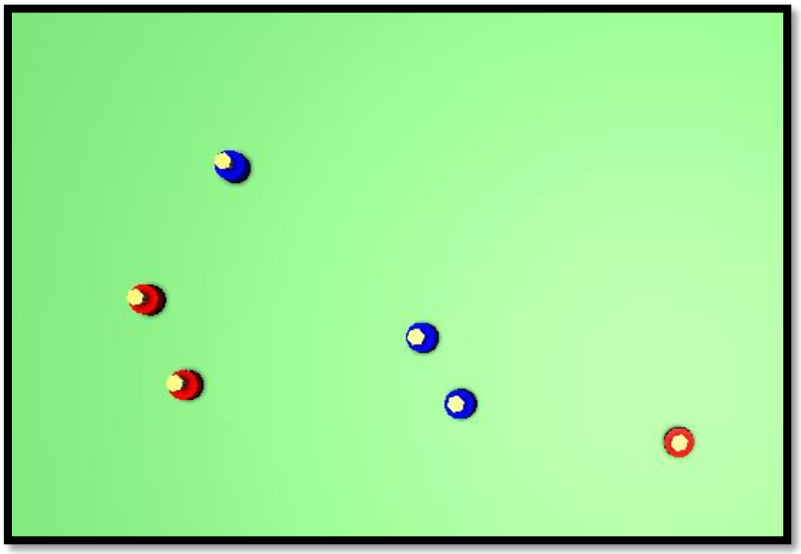

Fig. 9. : the view in day mode

4.4 RECRUITMENTS

Participants were recruited via the Sona platform, a system set up by ULB to allow their researchers to recruit participants for their experiment. First year psychology students must obtain 20 Sona credits to validate their year. Each proposed experiment has a certain credit value related to the duration of the experiment. Our online experiment was 20 minutes long and allowed students to obtain one credit.

Slots were opened on the Sona platform for a controlled number of students to enrol. To reach our data objective, we had to manage more than 340 participants online registrations for 180 effective participants. They had an average age of 20.54 years and were 82% female. The difference between registrations (340) and effective participants (180) was due to difficulties with online connection problems or the absenteeism of participants, which sometimes prevented from the opportunity to obtain an even number of participants. To optimise the process, we modified the code of our game to be able to add Zealots in a more flexible way, i.e. during the game rather than only in a fixed way at the beginning from the control panel shown in Fig. 10. This, and the possibility to reject participants, ensured that we consistently had an even number of participants, resulting in a total of 10 agents at all times.

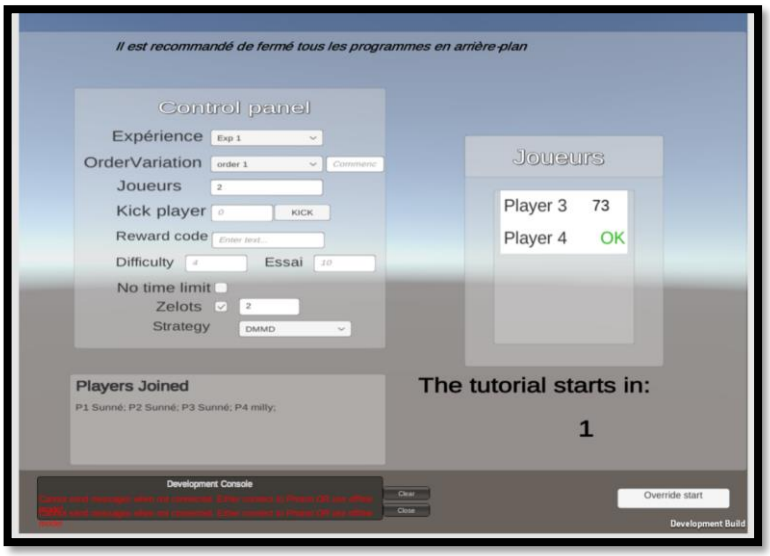

Fig. 10 : control panel showing the connected participants on the right and the configuration of the number of zealots and maximum number of participants on the left.

4. HYPOTHESIS

Building upon the insights gained from the literature review, this chapter presents the research question being addressed, followed by a set of well-defined hypotheses. In the analysis of the existing literature on collective decision-making, several gaps have been identified, alongside areas that merit further exploration, such as the study of human and artificial agent interactions and the incorporation of parameters found in human-human swarms, such as cognitive diversity. Additionally, the effects of zealots are not consistent, indicating that further investigation could provide valuable insights. The research question thus aims to contribute new knowledge to the field by drawing upon the key points from our literature review.

5.1. RESEARCH QUESTION

What is the impact of an increasing proportion of stubborn artificial agents named “zealots” on the decision-making process and outcomes in human groups through an online experiment. The online experiment involves each human participant in the group determining the dominant color on the bi-color online platform (red or blue).

5.2. FOUR HYPOTHESES

Based on our research problem, the following four hypotheses are proposed using the dependent variable described in this table.

Dependent Variable	***Definition***
The majority	The majority represents the proportion of individuals within the group who share the same opinion. The majority value ranges from 0.5 to 1, encompassing every tenth in between. A majority ratio of 0.5 indicates that the group is evenly divided, with half of the individuals holding one opinion (red or blue) and the other half holding the opposite opinion. In other words, there is no consensus among the group members. A majority of 1 indicates that all the group hold the same opinion. That possibility can happen only in the absence of zealots (Group 5).
The participant majority	The participant majority is the majority opinion among human participants. It can variate from 0.5, indicating that the participants are evenly divided, to 1, indicating that the whole group have the same opinion.
The correct majority	The correct majority represents the proportion of all agents within the group (humans and zealots) who share the same correct opinion. This value ranges from 0 to 0.4 if all the participants would have the same wrong opinion, 0.5 if the opinion is equally divided and from 0.6 to 1, if all participants would have the same right opinion.
The participant correct majority	The correct majority represents the proportion of human participants who share the same correct opinion. This value ranges from 0 to 0.4 if all the participants would have the same wrong opinion, 0.5 if the opinion is equally divided and from 0.6 to 1, if all participants would have the same correct opinion.
The accuracy	The accuracy can only take two outcomes, correct or incorrect. For the statistics, every ration under 0.5 was transformed in 0, greater than 0.5 in 1 and 0.5 was kept.

*5.2.1 As the percentage of zealots increases, the **majority** decreases.*

As the percentage of stubborn agents increases in a group, the number of elements in the group sharing the same decision decreases. With more zealots present, the group becomes more divided, leading to a lower majority value. For instance, with 8 zealots, the majority can be 0.5 (4 Zealots Red, 4 Zealots Blue, 1 human B and 1 human R) or 0.6 (4 Zealots Red, 4 Zealots Blue, 2 human B or R). With 2 Zealots and 8 humans, the majority can reach 0.9 and with no Zealots, the majority can reach 1.

*5.2.2 An increase in the percentage of zealots will lead to a decrease in the **participant majority**.*

Increasing the percentage of zealots would expose humans to a larger variety of opinions and would reduce common social information leading humans to lower and slower consensus. Hypothesis 2 is supported by the work of Vendeville et al. (59). Vendeville et al. (59) simulated a full graph at various times and demonstrated that if zealots are present in only one camp, the entire population will eventually adopt the opinion of the zealots' camp. However, if zealots are present on both sides, they prevent consensus, causing the system to reach a state of equilibrium where no single opinion prevails.

*5.2.3 An increase in the percentage of zealots is expected to result in a higher proportion of **correct majority** opinions.*

Hypotheses 3 is supported by the work of Prasetyo et al. (49) as illustrated in the literature review in the Robot Swarm section. The research performed a study of the best-of-n problem in dynamic environments with two alternatives available, the same as the current experiment. The decision mechanism they used is the voter model which looks like the way humans and artificial agent express their choice in the current experiment, i.e. the ability of switching color and switching from dissemination phase (day mode) to exploration phase (night mode) as explained in the methodology. The main outcome of Prasetyo et al.

(49) was that without stubborn individuals, the group did not adapt to a dynamic environment. They tested swarms of 100, 1000 and 10000 agents in the network with two fixed proportions of zealots (5% and 20%) and a fixed number of zealots per group (Prasetyo et al. (49)). A large number of stubborn agents had no effect or detrimental effects, but a small percentage of zealots could achieve a good level of consensus in swarm of 1000 and 10.000 agents. For practical reasons related to the presence of humans, the current experiment included 0% to 80% of zealots. Furthermore, a sub-hypothesis is that, there is a threshold number of zealots beyond which accuracy drops, as described by Prasetyo et al. (49).

*5.2.4 The number of **correct** decisions made by the human participant is expected to increase as the percentage of zealots grows.*

Mathematical and artificial models have primarily shown positive effects of incorporating zealots (Vendeville et al. (59), Prasetyo et al. (49)). However, when accounting for human and social interactions, the outcomes often become more negative, leading to polarization or backlash effects (Jones et al. (66)). Vendeville et al. (59) demonstrated that injecting zealots into a group can provoke a "back-fire" effect, causing adverse reactions and reinforcing polarization levels. Consequently, our hypothesis appears fragile. One potential argument in support of this hypothesis is the concept of cognitive diversity.

Cognitive diversity has been shown to be crucial for collective intelligence (Hong and Page 2001, 2004, 2009; Page 2007; Landemore 2012a, 2012b, 2013), as illustrated in the last section of the literature review. This diversity is a characteristic of the group as a whole rather than of individual members. In problem-solving situations, the quality of the collective answer relies more on cognitive diversity than on individual intelligence (Hong and Page 2001). In contexts where judgments are aggregated to a collective answer, Hong and Page's account, the reason why the aggregation of predictive models can do a better job of producing correct decisions comes from the existence of negative correlations between voters' predictions. Negative correlations between voters' predictions can result in

cancelling out individual mistakes and producing a smaller group average error. The inclusion of zealots, with their distinct perspectives and beliefs, could enhance cognitive diversity and ultimately improve decision-making. The assumption is that such an effect is more likely to occur in large group.

5. RESULTS

6.1 DATA COLLECTION

The experiment conducted with HuGos has enabled an exploration of the data recorded from 2160 minutes of an online task. A total of 177 2 minutes of trials from 5 types of groups was recorded from a total of 30 groups (6 groups consisting of each 5-of the following number of zealots: – 0, 2, 4, 6, 8) played a session of 6 times 2-minutes trials. Thus, the data set was constructed of N = 180 (30 groups x 6 trials) (60+48+36+16+12 = 172 participants). The task recorded a wide variety of variables with the number of zealots as independent variable and the following dependent variable: the majority, the correct majority, the participant majority, the participant correct majority and the accuracy.

6.2 STATISTICAL ANALYSYS OF HYPOTHESIS

All reported analyses were conducted using Python and RStudio. The impact of an increasing proportion of stubborn artificial agent on the decision-making process and outcomes in human groups in an online experiment is examined through the following four hypotheses.

*6.2.1 First hypothesis: As the percentage of **zealots** increases, the **majority*** decreases.

According to mathematic common sense, as the percentage of stubborn agent increases in a group, the number of elements in the group sharing the same decision decreases. With more zealots present, the group becomes more divided, leading to a lower majority value.

An analysis of variance (ANOVA) was performed to examine whether the group of 10 agents exhibit differences in their common choice based on the number of zealots, with the number of Zealots (5 levels : 0,2,4,6,8) as the independent variable and the average of the majority score as the dependent variable. A **significant p-value** (F (1, 35) = 44.44, p =1.03e-07 ***) was obtained. To visualise the data, a graph with plots was generated and it reveals the majority decreasing with the number of zealots (Fig. 11). Additionally, a mixed

model test was performed with the same independent variable and the number of zealots as fixed effect. A score of 71% (estimate = 0.719, p = < 2e-16 ***) of Majority on average was found when there are no zealots. This score decreases 2% per zealot present (estimate = -0.0208 p = 8.86e-08 ***).

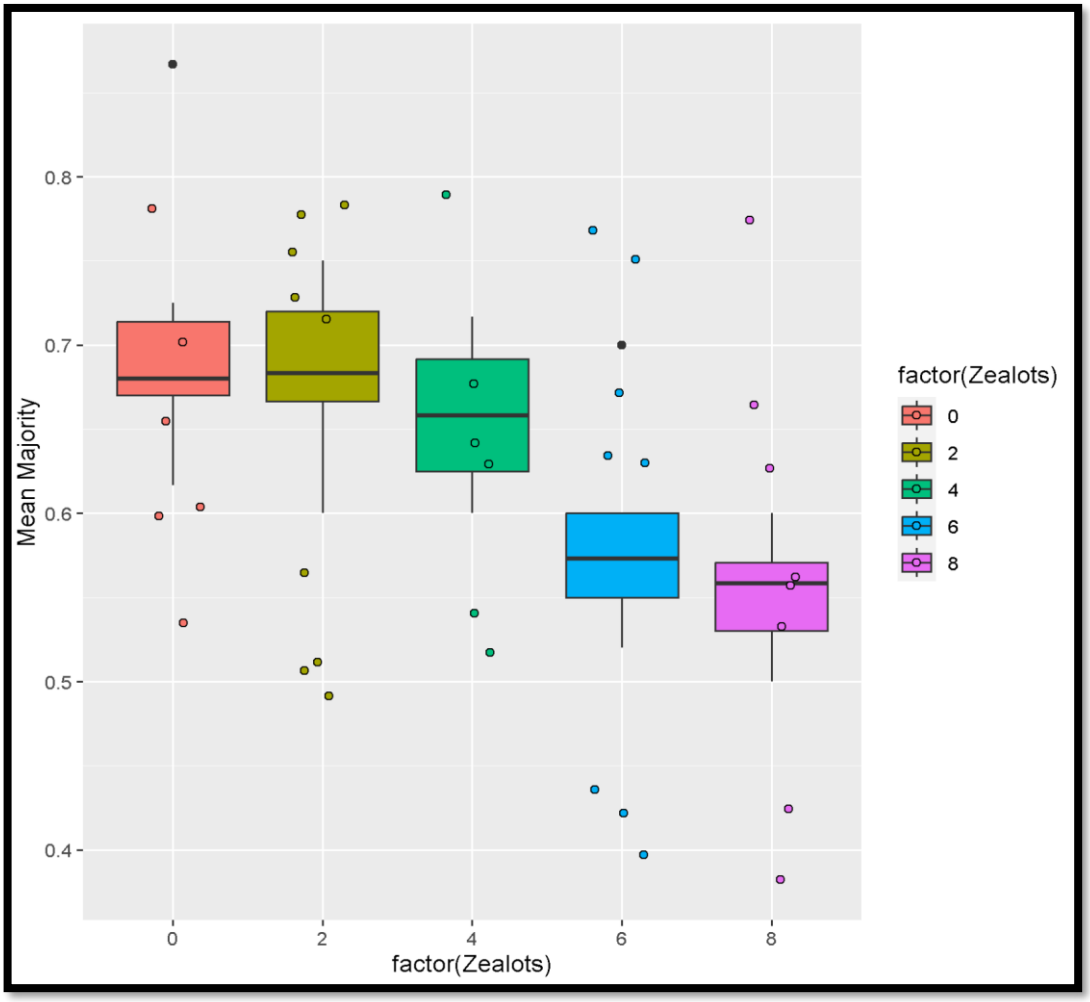

Fig. 11: Plots of the **average** Majority and number of zealots.

6.2.2 Second hypothesis: *An increase in the percentage of **zealots** will lead to a decrease in the **participant majority.***

To examine whether the human participants exhibit differences in their common choice based on the number of zealots, an analysis of variance (ANOVA) was performed with the number of Zealots (5 levels : 0,2,4,6,8) as the independent variable and the mean of participant majority per session as the dependent variable. A **non-significant** p-value ($F(1, 35) = 0.438$; $p = 0.512$) was obtained. To visualise the data, graph with plots was generated in which a mild tendency to decrease can be perceived (Fig. 12). Additionally, a mixed model test was performed with no significance.

Those results are supported by Vendeville et al.'s (59) study in which their model's strategy based on an arbitrary degree distribution network demonstrated consensus is prevented if zealots are present on both sides.

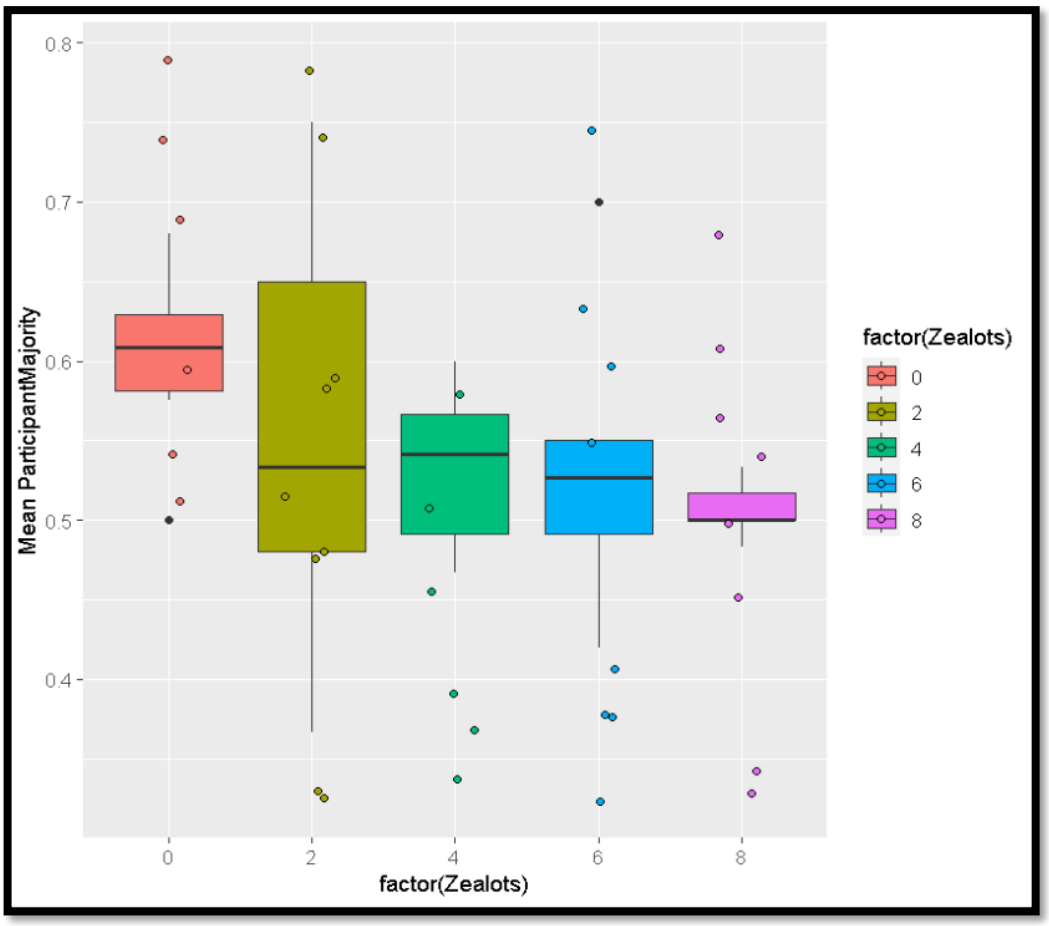

Fig. 12. : Plots of Mean Participant Majority and number of zealots.

*6.2.3 Third hypothesis: An increase in the percentage of **zealots** is expected to result in a higher proportion of **correct majority** opinions.*

To examine whether the human participants exhibit differences in their common choice based on the number of zealots, an analysis of variance (ANOVA) was performed with the number of Zealots (5 levels : 0,2,4,6,8) as the independent variable and the average correct majority per session of the group (humans and zealots) as the dependent variable. A **significant p-value** (F (1, 35) = 4.815, p=0.0349) was obtained. To visualise the data, a graph was generated in which a verry mild negative slope can be seen (Fig. 13). Additionally, a mixed model test was performed with the correct majority as the independent variable and the number of zealots as the fixed effect. A score of 60% (estimate = 0.588, p = < 2e-16 ***) of Majority was found on average when there are no zealots. This score decreases 1% per zealot (estimate = -0.0117 p = 0.008 **) as can be observed in Fig. 14 (unlike the previous graph, this one is not based on the mean and makes the visual clearer for the estimate).

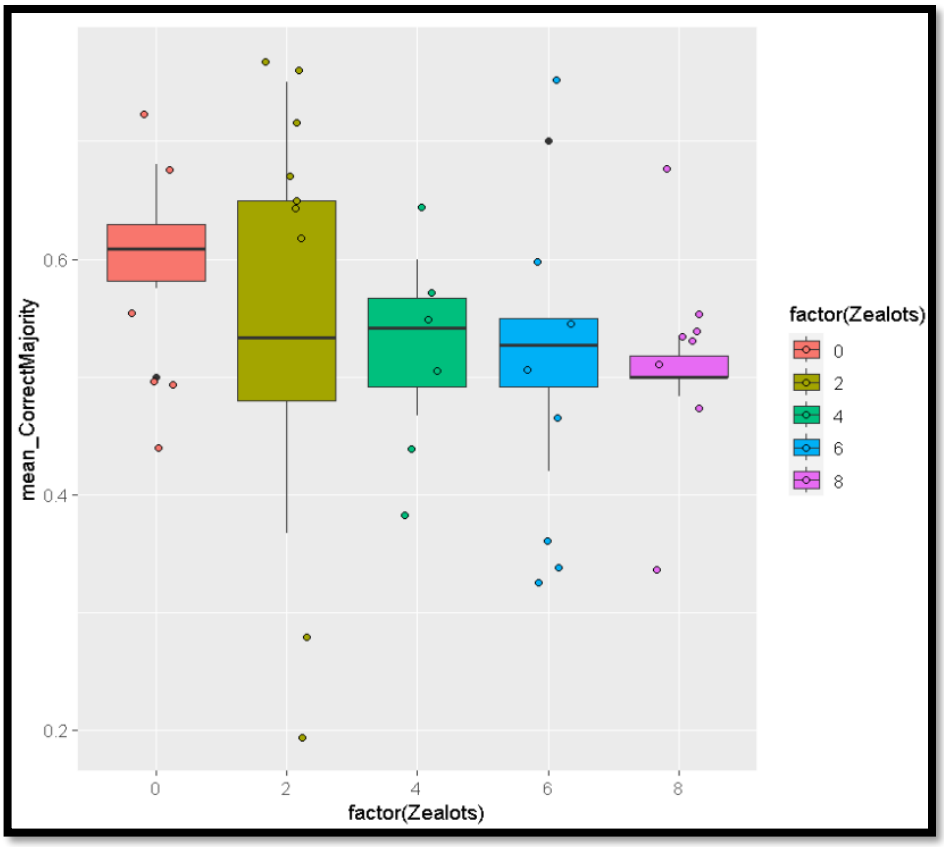

Fig. 13.: Plots of **Average** Correct Majority and number of zealots.

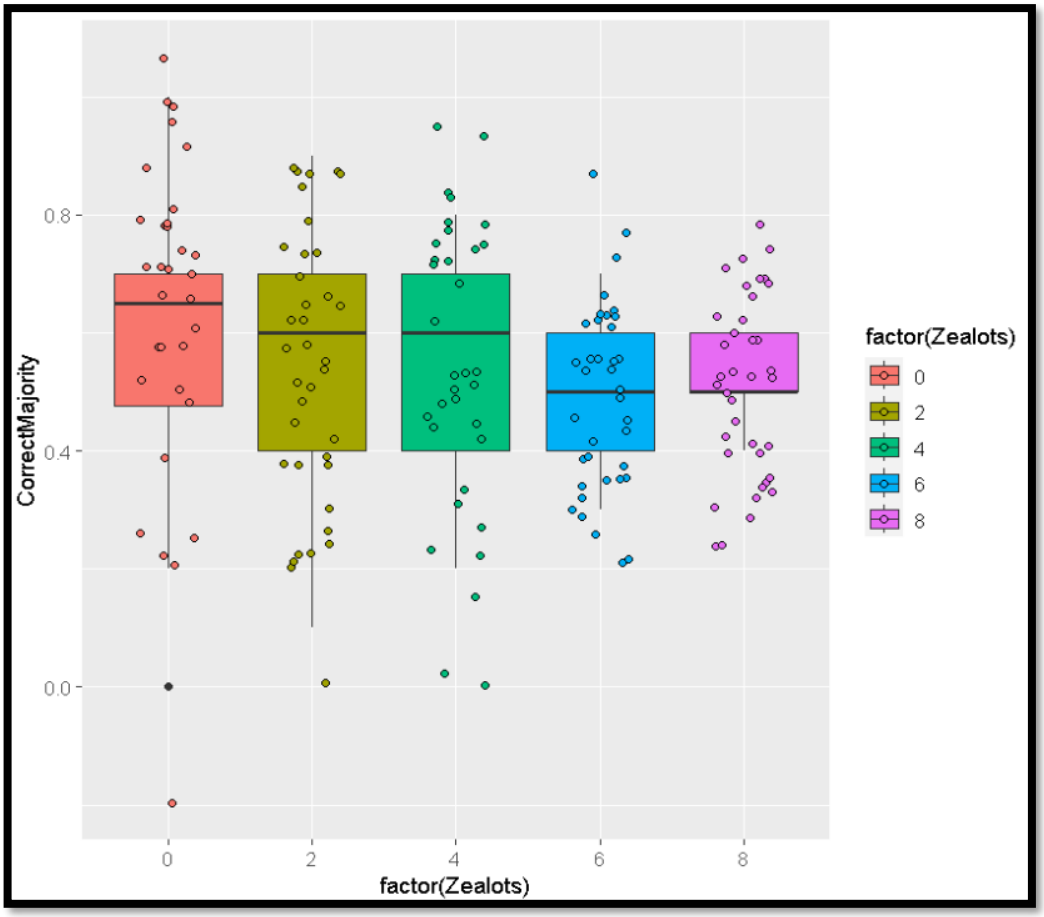

Fig. 14.: Plots of Correct Majority and number of zealots.

*6.2.4 Fourth hypothesis: The number of **correct decisions** made by the group is expected to increase as the percentage of **zealots** grows.*

To examine whether the accuracy of correct decisions made by the group exhibits differences based on the number of zealots, an analysis of variance (ANOVA) was performed with the number of Zealots (5 levels : 0,2,4,6,8) as the independent variable and the accuracy as the dependent variable. A **non-significant p-value** ($F(1, 35) = 2.099$, $p = 0.156$) was obtained. To visualise the data, a graph was generated in which a tendency for the plot to decline (Fig. 15) can be seen. This graph reveals that the spread of the condition with 2 zealots is particularly large, meaning a large part of the groups are always correct and a large part of the groups are always wrong (under 0.5). Additionally, a mixed model test was performed with the accuracy and the number of zealots as fixed effects. A

significant score of 67% (estimate = 0.678, p = < 2e-16 ***) of Majority on average was found when there are no zealots. This score decreases 2% per zealot (estimate = -0.022 p = 0.0548).

The presence of zealots mildly lowers the performance of the group. A possible reason for this comes from Vendeville et al. (59) who demonstrated that injecting zealots into a group can provoke a "back-fire" effect, causing adverse reactions.

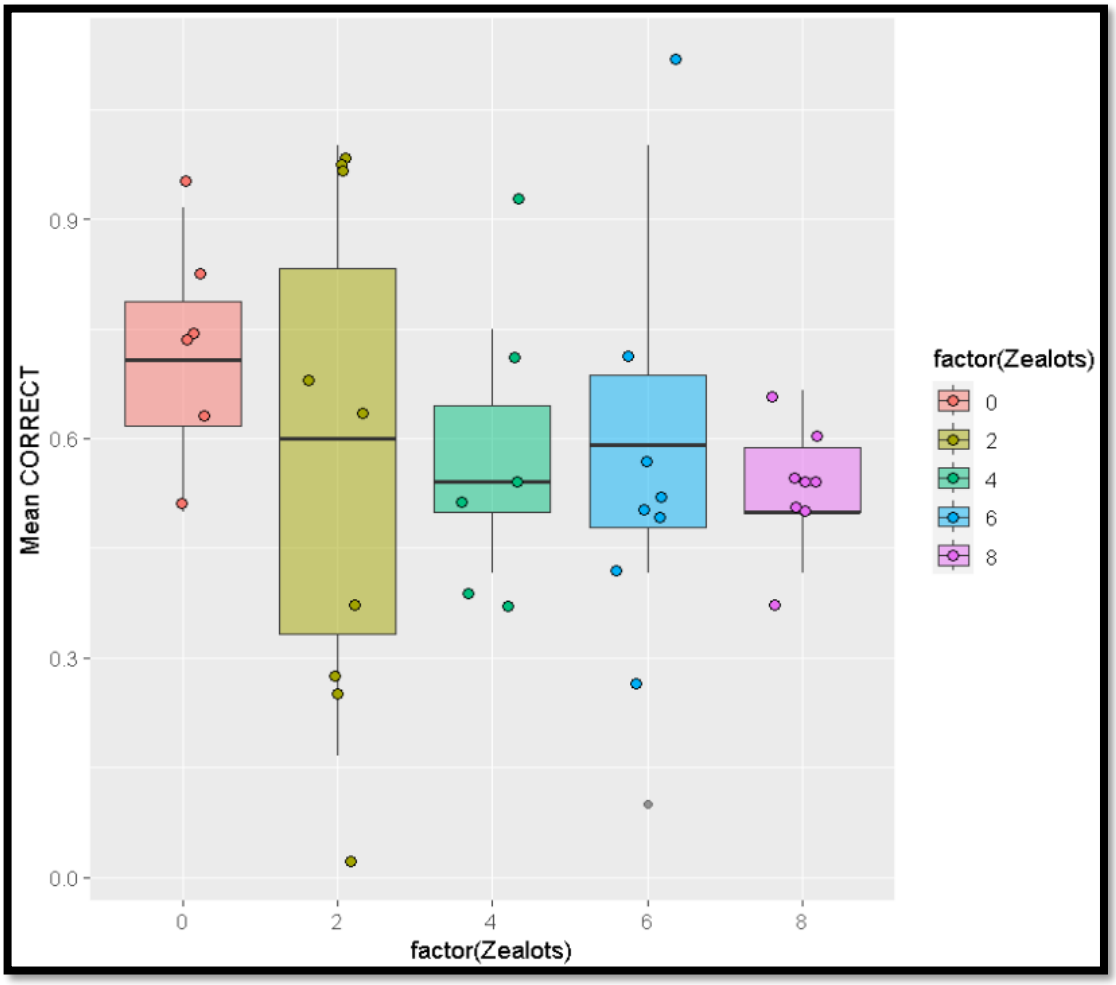

Fig. 15. : Plots of the accuracy and number of zealots.

Furthermore, out of our four hypotheses, the variable correct participant majority was explored and did not show significant results.

6. DISCUSSION

7.1 OVERVIEW OF THE RESULTS

The present study explored the impact of an increasing proportion of stubborn artificial agents, from 20% to 80% and a condition without zealots on the decision-making process and outcomes. Our analysis revealed the following key findings related to our four main hypotheses:

1. As the percentage of zealots increases, the majority decreases. Our results validate this hypothesis, as shown by a significant p-value obtained from an analysis of variance (ANOVA) and the negative relationship between zealots and majority in the mixed model analysis. Align with mathematical common sense, the presence of more zealots in the group leads to a more divided group, resulting in a lower majority value.

2. An increase in the percentage of zealots will lead to a decrease in the participant majority. Although this hypothesis is supported by previous research (Vendeville et al., 59), our statistical analysis did not show a significant relationship between the number of zealots and the participant majority. This suggests that further investigation is needed to understand the effect of zealots on participant majority in different contexts.

3. An increase in the percentage of zealots is expected to result in a higher proportion of correct majority opinions. Our statistical analysis revealed a significant relationship between the number of zealots and the correct majority of the group (humans and zealots) however, the correlation between the two variables was negative. This finding aligns with the work of Prasetyo et al. (49), who found that a large number of stubborn agents could have detrimental effects on the group consensus. Our study, however, did not identify the threshold number of zealots beyond which accuracy drops.

4. The number of correct decisions made by the group is expected to increase as the percentage of zealots grows. Contrary to our hypothesis, the presence of zealots mildly lowered the performance of the group with mild significant results obtained with a mixed model. Those results are consistent with Vendeville et al. (59), who found that injecting zealots into a group can provoke a "back-fire" effect, causing adverse reactions.

The findings highlight the complex relationship between the presence of zealots and group decision-making outcomes. While increasing the percentage of zealots may lead to a decrease in the majority and correct majority opinions, the impact on participant majority and accuracy remains uncertain. Further research is needed to explore the potential threshold number of zealots and the underlying mechanisms that drive the observed effects in group decision-making processes.

7.2. LMITATIONS OF THE STUDY AND FUTURE RESEARCH DIRECTION

One of the main limitations of this study lies in the size of the groups which were tested. Expanding the swarm size and reducing the proportion of zealots could be a path to explore. Prasetyo et al. (49) illustrate that the influence of zealots had more positive impacts in larger swarms. We can envision that the improvement of future virtual tools will allow for more stable and stronger connections to run online tasks with swarms of, for example, a hundred humans. This will provide interesting avenues for human decision-making and enhancing collaboration between robots, as these fields are interconnected and can learn from each other's advancements.

A second way to improve this study would be to add conditions such as the level of confidence, which would increase the complexity of the analysis, but provide deeper insights. While a relatively simple setup was intentionally used, through further time investment, it would be possible to increase the complexity by drawing inspiration from both social interaction and robotic technology axes. As seen in this research project, it is crucial to consider both of these axes to avoid generating artificial results disconnected from human reality. As Dahlgren et al. (70) points out, the impact of artificial technology and human societal perspectives should be differentiated. For instance, while some studies at the technological level find evidence of extremely large filter bubbles using mathematical models or simulations (Chitra & Musco (72), research incorporating human and societal perspectives typically finds little to no evidence for filter bubbles, and often even discovers the opposite (Zuiderveen Borgesius et al. (73), Bruns et al (74)).

For instance, the voting process could be enhanced by allowing participants to express their degree of confidence in their decision. A system could be designed in which participants input their confidence level with a 1 to 5 scale, which would be

displayed during the dissemination phase (day mode). Numerous studies have shown the influence of confidence and how it can lead to confirmation bias (Rollwage et al. (82), Kappes et al. (68)), while others have shown improved performance (Compte et al. (83)). Both paths could be explored. Additionally, this aligns with the more sophisticated voter model proposed by Valentini et al. (58) in the robotic technology field as this tool could be likened to the human expression of confidence level. This tool refers to the weighted voter model, where, for a time proportional to the quality of the option corresponding to their opinion, robots broadcast their choice to other robots within their range of sight. This concept is inspired by the waggle dance of honeybees (Prasetyo et al. (49)) highlighting the interconnectedness of the research area.

7.3. GENERALIZATION TO OTHER SCENARIOS AND SOCIETY

The research insights gained could potentially be extrapolated to a diverse range of other scenarios that extend beyond online decision-making experiments. These scenarios include social contexts, political environments, and social media platforms where the role of 'zealots' may be represented by individuals with unyielding and entrenched viewpoints.

For instance, a study conducted by Stewart et al. (81), developed a 'voter game' model to examine how information flows during collective decision-making, particularly in social contexts. Players in this model were assigned to competing groups and placed on an 'influence network,' which determined whose voting intentions they could observe (Stewart et al. (81)). The researchers discovered a phenomenon called 'information gerrymandering,' where the structure of the influence network can bias vote outcomes towards one party, even if both parties are equally sized and each player has the same influence. Strategically placed zealots could also trigger this bias. These findings were confirmed through social network experiments with 2,520 human subjects and observed in real-world scenarios, such as online political discussions before US federal elections and patterns of bill co-sponsorship in the US Congress and European legislatures (Stewart et al. (81)). The study highlights how collective decision-making can be vulnerable to distortion due to restricted information flow and potential deadlock when multiple parties engage in information gerrymandering (Stewart et al. (81)).

Continuing in a political environment, a recent study by the Pew Research Centre found that, despite significant events between 2018 and 2020 like the global pandemic, mass protests against racial injustice, and a presidential impeachment, the vast majority of both Republican and Democratic voters in the US retained their party affiliations (Jones et al. (66)). Only 9% of voters from each party changed their affiliations during this period. This finding, based on interviews with 11,077 registered voters, suggests that a sense of group identity or loyalty may lead voters to maintain their affiliations despite events that could potentially shift their political perspectives (Jones et al. (66)).

On social media platforms, the presence of zealots, in the form of bots or humans with fixed views, can influence the dynamics of online discussions and shape public opinion. This effect is particularly noticeable in an era where social media platforms' recommendation algorithms tend to foster polarization. These algorithms can exacerbate confirmation bias by mainly displaying materials that fit users' preferences, creating a "filter bubble" effect (Eli Pariser (18)). The "filter bubble" thesis posits that personalization algorithms on social networking sites create isolated information ecosystems that can reduce exposure to opposing viewpoints and lead to potentially detrimental consequences for democracy.

However, several limitations should be acknowledged when generalizing these findings. Firstly, the experimental context was a controlled online environment with a specific task (color identification task). This starkly contrasts with real-world scenarios like political debates or social context decision-making, where numerous choices can be offered. Secondly, the results from the current experiment are based on a limited sample size and a relatively simple decision-making model. Real-world scenarios often involve larger groups and more complex decision-making processes, which could significantly alter the dynamic observed in our study. Lastly, this study did not consider the potential impact of the source of zealotry. In real-world scenarios, the credibility, expertise, or charisma of the zealots could significantly influence their impact on group decisions, a factor not explored in this study.

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